The limit of this as $\Delta \mathrm{E} \to 0$ defines a quantity

$$\mu(\mathcal{M}) = \int_{\mathcal{M}} \frac{1}{\|\nabla H\|} \mathrm{d}\sigma.$$

As both volume and energy are invariant under the Hamiltonian flow, $\mu(\mathcal{M})$ must be invariant under the flow, thus

$$\mu(\mathscr{F}_t(\mathcal{M})) = \mu(\mathcal{M}).$$

The invariant measure of the whole level set Σ_{E} is defined as the *microcanonical partition function*

$$Z_{\mathrm{E}} = \mu(\Sigma_{\mathrm{E}}).$$

Example 5.1 Consider the harmonic oscillator $H(q,p) = p^2/2 + \Omega^2 q^2/2$. The energy surface $H = \mathrm{E} > 0$ is an ellipse. The partition function is obtained by integrating in the standard way. Parameterizing the surface by q, we have $p = \sqrt{2\mathrm{E} - \Omega^2 q^2}$, then

$$\begin{aligned}
\mu(\Sigma_{\mathrm{E}}) &= 4 \int_0^{\sqrt{2\mathrm{E}}} \frac{\sqrt{1 + p_q^2}}{\sqrt{H_p^2 + H_q^2}} \mathrm{d}q \\
&= 4 \int_0^{\sqrt{2\mathrm{E}}/\Omega} \frac{\sqrt{1 + \frac{\Omega^4 q^2}{2\mathrm{E} - \Omega^2 q^2}}}{\sqrt{p^2 + \Omega^4 q^2}} \mathrm{d}q \\
&= 4 \int_0^{\sqrt{2\mathrm{E}}/\Omega} \frac{1}{\sqrt{2\mathrm{E} - \Omega^2 q^2}} \mathrm{d}q \\
&= 2\pi/\Omega.
\end{aligned}$$

Example 5.2 Consider a system of N non-interacting particles of mass m moving in periodic boundary conditions within a box of volume V. The Hamiltonian is

$$H = \frac{1}{2m} \sum_{i=1}^{\mathrm{N}} \|\boldsymbol{p}_i\|^2$$

so

$$Z_{\mathrm{E}} = \mathrm{V}^{\mathrm{N}} \int_{\mathrm{S}} \frac{m}{r} \mathrm{d}\sigma$$

where $\mathcal{S}$ is the $3N-1$ dimensional sphere in $3N$ dimensions of radius $r=\sqrt{2mE}$, and $d\sigma$ is the surface area element on this sphere. V^N is the contribution due to integration over positions. Thus we have

$$\begin{aligned} Z_E &= V^N \frac{m}{\sqrt{2mE}} A(\mathcal{S}) = V^N \frac{\sqrt{m}}{\sqrt{2E}} 3N\pi^{3N/2} \frac{(2mE)^{(3N-1)/2}}{\Gamma(3N/2+1)}, \\ &= V^N \frac{(2\pi mE)^{3N/2}}{E\Gamma(3N/2)}, \end{aligned}$$

where Γ is the gamma function, and we have used the standard formula for the surface area of the $3N-1$ dimensional sphere.

It should be clear that, despite these simple examples, the invariant measure of a subset of phase space is rarely obtainable in an analytic form.

Note that in typical treatments of systems involving identical particles, the partition function is normally multiplied by a factor $\hbar^{-3N}/N!$ that measures the number of distinct quantum states available. The $N!$ quantifies the number of possible arrangements of N identical objects, since, in quantum mechanics, identical particles are formally indistinguishable. It would become complicated to incorporate such factors for a system with many different species of atoms. When computing probabilities, we would need to incorporate a corresponding counting factor which would, in typical cases, cancel with the prefactor. Our interest in this book is in classical systems of particles, and the number of particles is fixed in discussion of any particular system, so we may simply ignore this factor in our discussion.

5.3.3 The Microcanonical Probability Measure and Averages

Let

$$d\mu_E = Z_E^{-1} \frac{1}{\|\nabla H\|} d\sigma.$$

We refer to this as the microcanonical probability density and write, for an "appropriate" set $\mathcal{A}$,

$$\Pr(\mathcal{A}) = \mu_E(\mathcal{A}) = \int_{\mathcal{A}} d\mu_E.$$

The only requirement for this to make sense is that the set $\mathcal{A}$ be a measurable subset of the $H = E$ energy surface, i.e. that it be possible to compute its surface area.